Reg. No. | | | | | | | | | | |

[14CE663]

SIXTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

REMOTE SENSING AND ITS APPLICATIONS

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Answer any **FIVE** full Questions selecting atleast **TWO** from each part.
2. Missing data may be suitably assumed.

PART - A

1.	a.	What is remote sensing? Explain briefly with the help of neat sketch, the elements of remote sensing system.	06
	b.	Define spectral reflectance curve? Discuss with neat diagram, the spectral reflectance characteristics of vegetation and water.	08
2.	a.	What is meant by multispectral scanner? Describe with neat sketch, the function of along track scanner	08
	b.	Briefly explain the spectral resolution and temporal resolution of satellite image.	06
3.	a.	Write the sensor capability of IRS.	07
	b.	Explain the characteristics of ERS.	07
4.	a.	Derive the equation to determine the height of an object using relief displacement with neat diagram	10
	b.	Enlist the advantages of IR aerial photograph over panchromatic photographs	04

PART - B

5.	a.	Name and discuss the methods of data interpretation.	06
	b.	Identify and describe the elements of visual image interpretation with examples.	08
6.	a.	Describe with neat diagram, how the histogram shift adjustment technique aids in improving the atmospheric corrections?	05
	b.	Give a detail explanation with the help of neat diagram how the supervised image classification technique used in thematic information extraction.	09
7.	a.	Tabulate the USGS Land Use Land Cover (LULC) classification system for use with remote sensor data.	08
	b.	Summarize the significance of remote sensing technique in mapping water quality.	06
8.	a.	Write short notes on proximity analysis and overlay analysis.	08
	b.	Define GPS? Enlist the application of GPS in civil engineering field.	06

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